

Euclid's Elements — Book XI

Proposition 10

Formal Reconstruction — Technical Documentation

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If two straight lines meeting one another be parallel to two straight lines meeting one another not in the same plane, they will contain equal angles.

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1 Introduction

Proposition XI.10 transfers an angle from one plane to another by means of two pairs of corresponding parallel lines. In Euclid’s configuration, AB and BC meet at B , while DE and EF meet at E . The first pair is parallel to the second pair, and the two angle configurations are not all contained in one plane. The conclusion is

$$\angle ABC \cong \angle DEF.$$

The proposition is easy to underestimate. Ordinary, unoriented parallelism is not sufficient to determine the correct angle. Reversing exactly one of the four rays changes the desired angle to its supplement. The formal reconstruction therefore has to retain directional information which Euclid leaves to the diagram and to the phrase “parallel to” in a particular direction.

The production proof remains fully synthetic. It uses no coordinates, vectors, scalar product, normal vectors, trigonometric functions, or numerical angle measure. Its central ingredients are:

- Euclid I.33 in explicitly chosen plane slices;
- Euclid XI.9 for spatial parallelism;
- a plane-separation / section-line bridge;

- spatial SSS;
- a final SameRay transport used by the public wrapper.

The proof also illustrates a distinction that is especially important in Book XI. The note intentionally contains several schematic figures: a main spatial overview, the I.33–XI.9 connector state, the directed-parallel slice, the section-line / plane-side bridge, the public-wrapper normalization, and the final SameRay/SSS states. Each figure is generated from an auditable Asymptote source.

unoriented carrier parallelism $\neq$ directed parallel segments.

The former is enough for XI.9. The latter is needed to recover exactly the angle intended by XI.10.

2 Euclid’s Statement

Euclid’s Proposition XI.10 states:

If two straight lines meeting one another be parallel to two straight lines meeting one another not in the same plane, they will contain equal angles.

Let AB and BC meet at B , and let DE and EF meet at E . Assume that the corresponding straight lines are parallel and that the two intersecting pairs are not contained in one common plane. Euclid’s claim is

$$\angle ABC = \angle DEF.$$

The phrase “parallel to” here carries directional content. If one ray is reversed while the other three are kept fixed, the conclusion is generally not equality of the two displayed angles. This directional issue is part of the mathematical statement, not merely a formalization detail.

3 Classical Configuration

Euclid first cuts four equal segments:

$$BA = BC = ED = EF.$$

He then joins

$$AD, \quad CF, \quad BE, \quad AC, \quad DF.$$

The resulting configuration contains three successive parallelogram-type steps.

First,

$$BA \parallel ED, \quad BA = ED$$

implies by I.33 that

$$AD \parallel BE, \quad AD = BE.$$

Second,

$$BC \parallel EF, \quad BC = EF$$

implies, again by I.33,

$$CF \parallel BE, \quad CF = BE.$$

Third, XI.9 gives

$$AD \parallel CF,$$

and equality follows through the common segment BE :

$$AD = CF.$$

A final use of I.33 yields

$$AC \parallel DF, \quad AC = DF.$$

The triangles

$$\triangle ABC, \quad \triangle DEF$$

therefore have three corresponding equal sides, and I.8 gives

$$\angle ABC = \angle DEF.$$

$$BA = BC = ED = EF$$

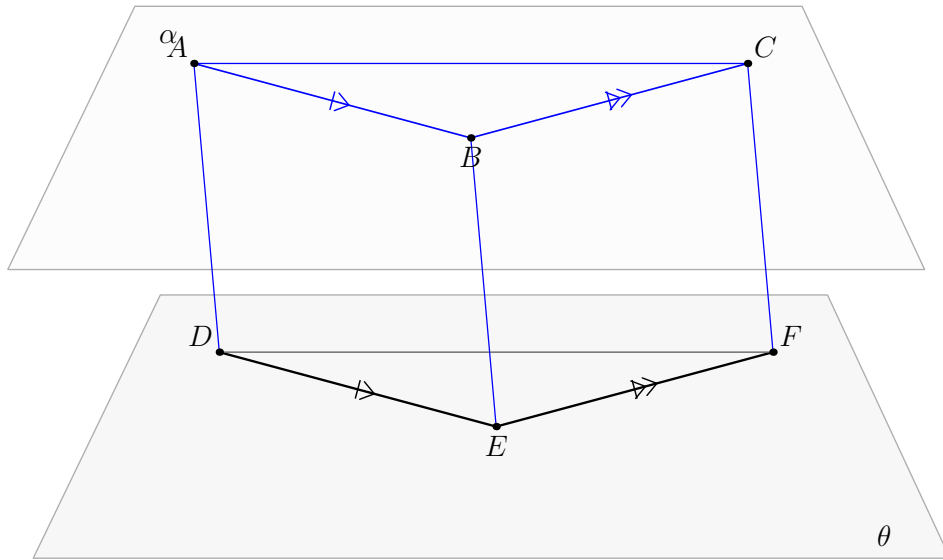


Figure 1: Classical spatial configuration for Proposition XI.10. The two lines meeting at B are compared with the two lines meeting at E ; the two angle configurations are not contained in one common plane. This figure is the global reference picture for the proposition.

4 Classical Proof

The classical local chain is

$$\begin{array}{c}
BA = BC = ED = EF \\
\Downarrow \\
BA \parallel ED, \quad BC \parallel EF \\
\Downarrow \\
\text{I.33 twice} \\
\Downarrow \\
AD \parallel BE, \quad AD = BE, \\
CF \parallel BE, \quad CF = BE \\
\Downarrow \\
\text{XI.9} \\
\Downarrow \\
AD \parallel CF, \quad AD = CF \\
\Downarrow \\
\text{I.33} \\
\Downarrow \\
AC \parallel DF, \quad AC = DF \\
\Downarrow \\
\text{I.8} \\
\Downarrow \\
\angle ABC = \angle DEF.
\end{array}$$

Thus the explicit classical proposition dependencies are

$$\text{I.3}, \quad \text{I.33}, \quad \text{XI.9}, \quad \text{I.8}.$$

The first step, I.3, is a normalization: equal pieces are laid off on the four rays. The geometric core begins after those equal segments have been chosen.

5 Why ordinary spatial parallelism is not enough

The project definition

```
HilbertSpaceLinesParallel Geo l m
```

records that two carrier lines are coplanar and disjoint. This is exactly the correct spatial analogue of ordinary line parallelism: skew disjoint lines are excluded.

XI.10 needs more. I.33 is directional: if two equal parallel segments are taken in corresponding directions, then the joining segments are equal and parallel in the corresponding direction. The same carriers with one endpoint order reversed describe the supplementary configuration.

For this reason XI.10 introduces

```
def HilbertSpaceDirectedParallelSegments
  [H : HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  (A B C D : Geo.Point) : Prop :=
```

```

exists pi : S.Plane,
exists l m t : Geo.Line,
  H.OnLine A l /\
  H.OnLine B l /\
  H.OnLine C m /\
  H.OnLine D m /\
  HilbertLineInPlane Geo l pi /\
  HilbertLineInPlane Geo m pi /\
  HilbertLinesDisjoint Geo l m /\
  H.OnLine B t /\
  H.OnLine D t /\
  HilbertLineInPlane Geo t pi /\
  HilbertSameSideInPlane Geo A C t pi

```

The ordered segments are $A \rightarrow B$ and $C \rightarrow D$. The connector t passes through the terminal points B, D . The initial points A, C must lie on the same side of that connector in the common plane of the two parallel carriers.

This is a synthetic orientation condition. It contains no coordinates and no selected global orientation of space.

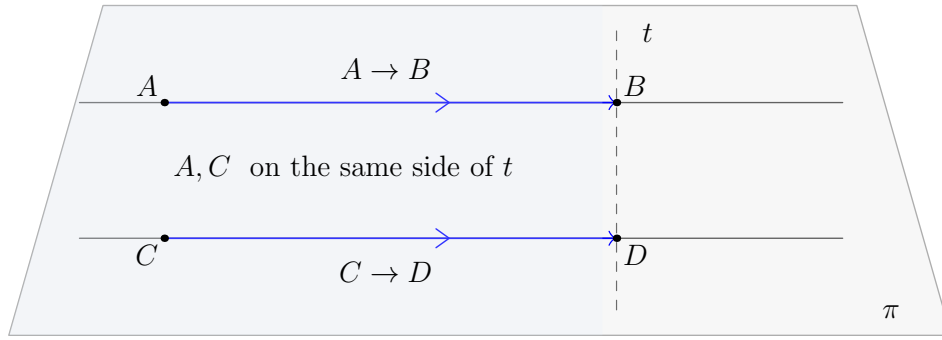


Figure 2: Directed parallel segments in a common plane slice. The carriers are parallel, the connector joins the terminal endpoints, and the initial endpoints lie on the same side of the connector. This is the synthetic orientation datum encoded by `HilbertSpaceDirectedParallelSegments`.

6 The normalized XI.10 core

The proof was first closed in a normalized form, in which the four selected arms already have the equal lengths used by Euclid:

$$BA \cong ED, \quad BC \cong EF.$$

At this stage the first two I.33 applications are represented by

```
hilbert_XI10_I33_directed_output
```

and packaged by

```
hilbert_XI10_first_two_I33_directed_package
```

The package supplies explicit carriers AD , CF , and BE , together with

$$\begin{aligned} AD &\parallel BE, & AD &\cong BE, \\ CF &\parallel BE, & CF &\cong BE, \end{aligned}$$

and, crucially, the directed versions

$$BE \longrightarrow AD, \quad BE \longrightarrow CF.$$

The ordinary carrier information is enough for XI.9. The directed information is retained for the later orientation step.

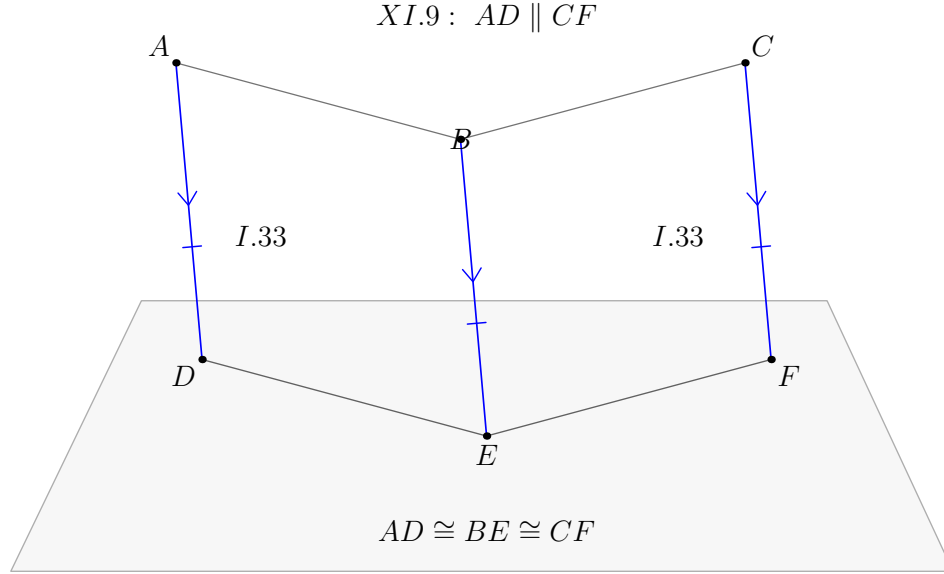


Figure 3: The connector state after the first two I.33 applications. The three joining segments AD , BE , and CF are congruent, while XI.9 supplies the carrier relation $AD \parallel CF$. This is the ordinary-parallel part of the normalized core.

7 The XI.9 step

The two first I.33 applications give two lines parallel to the same line BE . The hypothesis that the six selected points do not belong to one common plane supplies the noncoplanarity clause needed by XI.9.

Thus

```
euclid_proposition_11_9
```

gives

$$AD \parallel CF.$$

Segment congruence transitivity through BE gives

$$AD \cong CF.$$

At this point the unoriented part of the classical proof has been recovered. What remains is to prove that AD and CF have the correct *relative direction* for the third I.33 application.

8 Plane separation and the section-line bridge

This is the main formal point which is almost invisible in Euclid’s printed proof.

Let θ be the plane of the second angle, containing D, E, F . The first two directed I.33 steps imply, in two auxiliary planes, that A and B , and respectively B and C , lie on the appropriate sides of the section lines ED and EF .

To combine these data in ambient space the proof introduces:

```
def HilbertSegmentMeetsPlane ...
def HilbertSameSideOfPlane ...
```

The essential bridge is

```
hilbert_XI10_borsuk_szmielew_52
```

Its content is the following. Suppose two planes π, θ meet along a line t , and $P, Q \in \pi$. Then the segment PQ crosses θ exactly when it crosses the section line t inside π . Equivalently, under the usual off-boundary conditions, same-side in the plane π with respect to t is the same as same-half-space in ambient space with respect to θ .

The source-level identification of this bridge is collected in the Book XI source appendices. Here only its formal geometric role in XI.10 is needed.

No topological separation axiom is added. The bridge is derived from the existing spatial incidence and order structure.

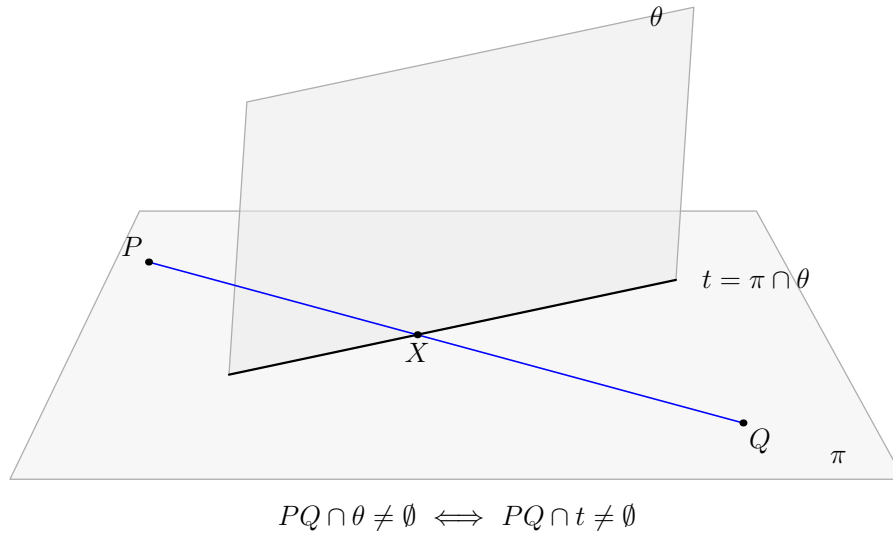


Figure 4: A segment PQ in a plane slice crosses the ambient plane θ exactly when it crosses the section line $t = \pi \cap \theta$ inside the slice plane π . This is the geometric bridge between plane-side and half-space information.

9 Transitivity of same side of a plane

The next helper is

```
hilbert_XI10_sameSideOfPlane_trans_noncollinear
```

It proves the transitivity needed for three noncollinear points:

$$A \sim_{\theta} B, \quad B \sim_{\theta} C \implies A \sim_{\theta} C.$$

The proof has two geometric cases.

If the plane $\alpha = \text{plane}(A, B, C)$ meets θ , their intersection is a line. The section-line / plane-side bridge converts both half-space relations into ordinary same-side relations in α ; the planar same-side paths compose there, and the result is converted back.

If α and θ have no common point, every point of the open segment AC remains in α , so that segment cannot meet θ . Hence A and C are automatically on the same side of θ .

This disjoint-plane branch is essential. One cannot assume that the two angle planes intersect.

10 Recovering the direction $AD \rightarrow CF$

The central orientation theorem is

```
hilbert_XI10_directed_AD_CF_from_common_BE
```

It combines

$$BE \longrightarrow AD, \quad BE \longrightarrow CF,$$

the ordinary conclusion

$$AD \parallel CF,$$

and the noncoplanarity of the two angle configurations.

The proof lifts the two local same-side statements to same-side-of-plane relations with respect to $\theta = \text{plane}(D, E, F)$, composes them there, and then cuts the result back down in the plane containing AD and CF . The section line is DF .

The conclusion is the exact directed statement

$$AD \longrightarrow CF.$$

This is the missing orientation datum which ordinary XI.9 cannot provide.

11 The third I.33 and spatial SSS

Now both ingredients required by the third I.33 are available:

$$AD \longrightarrow CF, \quad AD \cong CF.$$

The helper

```
hilbert_XI10_third_I33_AC_DF
```

applies the directed I.33 package and obtains

$$AC \cong DF.$$

The three side congruences are therefore

$$BA \cong ED, \quad BC \cong EF, \quad AC \cong DF.$$

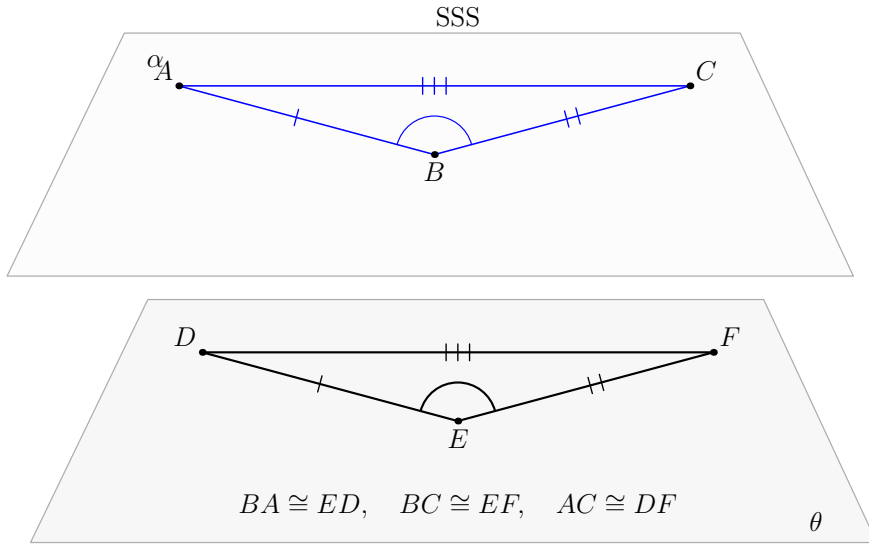


Figure 5: The final normalized SSS state. The three pairs of corresponding sides are marked separately: $BA \cong ED$, $BC \cong EF$, and $AC \cong DF$. The two triangles lie in distinct parallel plane slices.

The final normalized step is spatial SSS:

```
hilbert_space_sss_angleA_in_plane
```

Applied to the triangles BAC and EDF , it yields exactly

$$\angle ABC \cong \angle DEF.$$

The target triangle is represented inside the explicit slice $\text{PlaneGeo}(\text{plane}(D, E, F))$, while the first triangle remains ambient. The SSS bridge is designed precisely for this cross-plane comparison.

12 The actual Lean DAG

The following diagram is intentionally kept in ASCII form. It is the actual proof spine of the normalized Lean reconstruction and should not be replaced by a cosmetically simplified diagram.

```
BA ~= ED, BC ~= EF
directed parallels
```

```

      |
    I.33 twice
      |
    AD ~= BE, CF ~= BE
      |
    XI.9
      |
    AD || CF
      |
  Borsuk-Szmielew 52
      |
    AD directed-parallel CF
      |
    I.33
      |
    AC ~= DF
      |
    spatial SSS
      |
    angle ABC ~= angle DEF

```

This DAG is not merely a restatement of Euclid’s local theorem references. The additional section-line / plane-side stage is the formal replacement for directional plane-separation information that is carried silently by the classical spatial diagram.

13 Fixed-connector normalization

For the public theorem a slightly different endpoint orientation is more convenient:

$$AB \longrightarrow DE, \quad CB \longrightarrow FE.$$

This is the theorem

```
euclid_proposition_11_10_normalized_fixed_connector
```

The advantage is that the connector in both directed-parallel relations is the same fixed line BE . When the outer endpoints are moved along their rays during segment construction, BE does not change.

Two small transport results support this form:

```

hilbert_XI10_directedParallelSegments_swap_pairs
hilbert_XI10_directedParallelSegments_transport_first_endpoints

```

This is a representation choice only. It does not change the geometry of Euclid’s proposition.

14 The public wrapper: Euclid’s I.3 normalization

The public theorem does not assume equal selected arms.

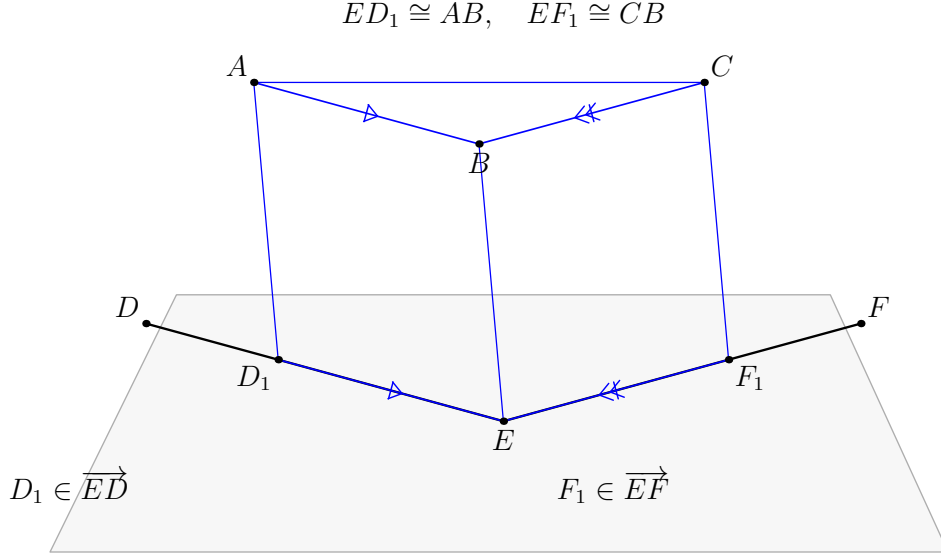


Figure 6: Wrapper-level segment construction. Only the points D_1 and F_1 are laid off on the rays ED and EF , with $ED_1 \cong AB$ and $EF_1 \cong CB$. This yields the normalized configuration used by the internal core theorem.

Instead, spatial segment construction lays off on the rays ED and EF new points D_1, F_1 such that

$$ED_1 \cong AB, \quad EF_1 \cong CB.$$

Only the second angle is normalized; the points A, B, C are left unchanged. This is formally more economical than reproducing all four layoffs of the classical prose.

Because

$$\text{SameRay}(E, D, D_1), \quad \text{SameRay}(E, F, F_1),$$

the directed parallel hypotheses transport to

$$AB \longrightarrow D_1E, \quad CB \longrightarrow F_1E.$$

The fixed-connector normalized theorem gives

$$\angle ABC \cong \angle D_1EF_1.$$

Finally, the SameRay angle bridge is applied inside the genuine plane $\text{plane}(D, E, F)$. This is an important architectural point: the proof does *not* install a global

[HilbertOrder Geo]

on ambient three-space. The ray-angle transport occurs legally in the induced planar geometry and is then mapped back to ambient angle data.

Thus

$$\angle D_1EF_1 = \angle DEF,$$

and the public conclusion follows.

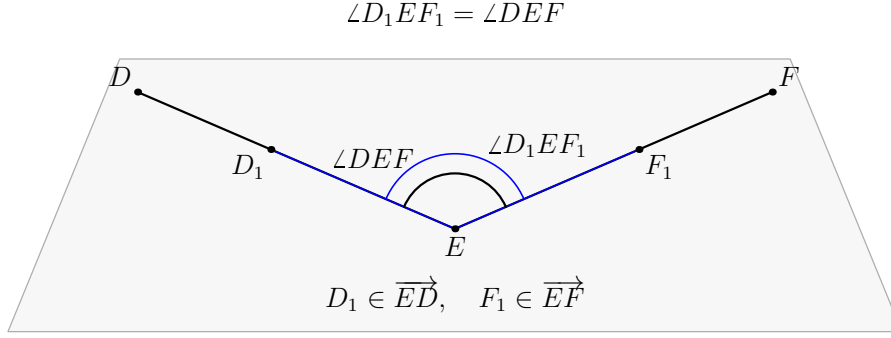


Figure 7: Final SameRay angle transport in the plane of DEF . After the normalized proof gives $\angle ABC \cong \angle D_1EF_1$, the facts that D_1 lies on the ray ED and F_1 lies on the ray EF identify the copied angle with the original target angle $\angle DEF$.

15 Formal Statement

The production theorem is:

```

theorem euclid_proposition_11_10
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  [HSE : HilbertSpaceEuclidean Geo]
  (A B C D E F : Geo.Point)
  (hDirAB_DE :
    HilbertSpaceDirectedParallelSegments Geo A B D E)
  (hDirCB_FE :
    HilbertSpaceDirectedParallelSegments Geo C B F E)
  (hABC : Not (PrimCollinear Geo A B C))
  (hDEF : Not (PrimCollinear Geo D E F))
  (hNoCommonPlane :
    Not (exists omega : S.Plane,
      S.OnPlane A omega /\
      S.OnPlane B omega /\
      S.OnPlane C omega /\
      S.OnPlane D omega /\
      S.OnPlane E omega /\
      S.OnPlane F omega)) :
    Geo.AngleCongruent A B C D E F

```

The hypotheses separate four kinds of information.

1. `hDirAB_DE` and `hDirCB_FE` encode the two corresponding directed parallel pairs.
2. `hABC` and `hDEF` ensure that both displayed angles are nondegenerate.
3. `hNoCommonPlane` states the genuinely spatial condition that the six selected points do not all belong to one plane.

4. The Hilbert typeclass hypotheses provide ambient incidence, spatial order, congruence, and plane-local Euclidean parallel uniqueness.

The conclusion is directly:

```
Geo.AngleCongruent A B C D E F
```

16 Spatial / Planar Architecture Used

The ambient layer is:

```
HilbertIncidence  
HilbertPlaneIncidence  
HilbertSpacePrimitive  
HilbertSpaceIncidence  
HilbertSpaceOrder  
HilbertSpaceCongruence  
HilbertSpaceEuclidean
```

The proof never assumes

```
HilbertOrder Geo  
HilbertCongruence Geo
```

globally on ambient space.

Whenever an ordinary planar theorem is needed, an explicit plane slice is used:

```
PlanePoint Geo pi  
PlaneLine Geo pi  
PlaneGeo Geo pi
```

The principal bridges used by XI.10 include:

```
planeGeo_sameSide_iff_space  
planeGeo_sameRay_iff_ambient  
planeGeo_angleCongruent_iff_ambient  
planeGeo_primCollinear_to_ambient  
planeGeoAngleToAmbient_angle  
hilbert_onPlane_of_primCollinear_with_two_on_plane  
hilbert_plane_intersection_line
```

The formalization therefore alternates between ambient three-space and ordinary planar Hilbert geometry, but it never conflates the two.

17 Hidden Formal Data

The classical diagram suppresses several obligations which are explicit in Lean.

17.1 Direction on each parallel pair

Carrier parallelism does not determine whether the selected ordered segments point in corresponding directions. XI.10 must store this information explicitly.

17.2 The connector line

Each directed parallel pair contains an explicit connector through the terminal endpoints. For the fixed-connector wrapper this line is BE .

17.3 The planes of the parallel pairs

The common plane of each pair of spatial parallels is an explicit witness. Disjointness alone would also admit skew lines and is therefore insufficient.

17.4 The plane of the second angle

The points D, E, F determine a genuine plane because

$$\neg \text{PrimCollinear}(D, E, F).$$

This plane is needed both for the half-space argument and for the final SameRay angle transport.

17.5 The intersection line in the section bridge

When two auxiliary planes meet, their common line must be constructed by

```
hilbert_plane_intersection_line
```

and characterized pointwise. It is never read off from a perspective drawing.

17.6 Disjoint angle planes are possible

The proof of same-side-of-plane transitivity cannot assume that the two angle planes meet. Write

$$\alpha = \text{plane}(A, B, C), \quad \theta = \text{plane}(D, E, F).$$

The branch $\alpha \cap \theta = \emptyset$ is a genuine part of the formal proof.

17.7 Noncoplanarity after segment layoff

The public wrapper replaces D, F by D_1, F_1 on the same rays. It must prove that a common plane containing A, B, C, D_1, E, F_1 would also contain the original D, F , contradicting the input hypothesis.

17.8 No ambient planar order instance

Several old SameRay utilities require `HilbertOrder Geo`. XI.10 does not introduce such an instance. The final ray-angle transport is performed inside the plane DEF , where the induced `PlaneGeo` legitimately has planar Hilbert order.

18 Relationship with Earlier Propositions

The direct Book XI dependency is XI.9:

$$AD \parallel BE, \quad CF \parallel BE \implies AD \parallel CF.$$

The proof also uses I.33 three times in mathematical substance:

1. BA, ED produce AD, BE ;
2. BC, EF produce CF, BE ;
3. AD, CF produce AC, DF .

The final triangle argument is I.8 / spatial SSS.

Euclid explicitly starts with I.3. In the current public Lean wrapper this role is played directly by

```
HilbertSpaceCongruence.segment_construction
```

rather than by importing a numbered I.3 theorem.

Thus there are two useful dependency descriptions:

Euclid's local DAG	current Lean route
I.3	<code>segment_construction</code>
I.33	directed I.33 helper
XI.9	<code>euclid_proposition_11_9</code>
I.33	directed I.33 helper
I.8	<code>hilbert_space_sss_angleA_in_plane</code>

The additional plane-separation stage has no separate Euclid proposition number. It exposes the section-line content needed to make the directional I.33 chain formally valid.

19 Axiomatic Strength

The production theorem assumes

```
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
```


in addition to the spatial incidence hierarchy.

The Euclidean layer enters because I.33 and XI.9 depend on Euclidean parallel uniqueness. No continuity principle is used.

The validated axiom audit is:

```
'Geometry.euclid_proposition_11_10' depends on axioms:
[proptext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

In particular:

- no new XI.10-specific axiom is introduced;
- no new global spatial separation axiom is introduced;
- no continuity axiom appears;
- no coordinate or analytic representation appears;
- bookZero_nullSegment1 is inherited project infrastructure.

20 Dependency Summary

The classical proof spine is:

```
I.3
|
I.33 twice
|
XI.9
|
I.33
|
I.8
|
angle ABC = angle DEF
```

The actual normalized Lean spine is:

```
BA ~= ED, BC ~= EF
directed parallels
|
I.33 twice
|
AD ~= BE, CF ~= BE
|
XI.9
|
AD || CF
```

```

      |
Borsuk-Szmielew 52
      |
AD directed-parallel CF
      |
I.33
      |
AC ~= DF
      |
spatial SSS
      |
angle ABC ~= angle DEF

```

The public-wrapper layer is:

```

original rays ED, EF
      |
segment construction
      |
ED1 ~= AB
EF1 ~= CB
      |
SameRay transport
      |
AB -> D1E
CB -> F1E
      |
normalized XI.10
      |
angle ABC ~= angle D1EF1
      |
SameRay angle transport
      |
angle ABC ~= angle DEF

```

These three diagrams should be read together. The first is Euclid's local theorem DAG. The second is the mathematical core actually realized by the Lean reconstruction. The third records the normalization wrapper needed to expose a public theorem with no prior equal-length hypotheses.

21 Conclusion

Proposition XI.10 is the first proposition in this part of Book XI where orientation of spatial parallels becomes unavoidable. Ordinary coplanar-disjoint parallelism suffices for XI.9, but XI.10 must remember which rays correspond. The dedicated `HilbertSpaceDirectedParallelSegments` relation supplies exactly that missing information without adding coordinates or a global orientation of space.

The formal proof preserves Euclid's structure: two I.33 steps create a common parallel BE , XI.9 compares the opposite joining segments, a third I.33 creates the third side equality, and SSS closes the angle comparison. The additional plane-separation bridge explains rigorously how the directional information passes between the relevant plane slices.

The production theorem compiles, the full project build is clean, and the axiom audit is

```
[propext, Classical.choice,  
  Geometry.bookZero_nullSegment1, Quot.sound]
```

with no new spatial or continuity axiom.